Umair Ejaz

AI / Machine Learning Engineer | Computer Vision | Embedded AI | Deep Learning

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EDUCATION

National University of Sciences and Technology — Islamabad, Pakistan

Bachelors in Electrical Engineering | CGPA: 3.56/4.00 — *Nov 2021 – Jun 2025*

PROFESSIONAL EXPERIENCE

MITACS CANADA — June 2024 – Sept 2024

Globalink Research Intern @ Toronto Metropolitan University — *Toronto, Canada*

- Designed and developed a fast motor controller board for precise current control in brushed and brushless DC motors, ensuring stable haptic interaction in virtual environments.
- Integrated USB 3.0, ADC/DAC converters, and a microcontroller for high-speed data acquisition, optimizing system performance with a 1kHz sampling frequency.
- Conducted circuit board design, testing, and prototyping to enhance accuracy, stability, and reliability, overcoming limitations of commercial motor driver systems.

UAS & Payloads Design Engineer — Jun 2025 – Feb 2026

NASTP — *Rawalpindi, Pakistan*

- Designed embedded control and sensor fusion systems for UAV payload stabilization and real-time telemetry, integrating signal processing and data analysis.
- Collaborated in hardware-software co-design to develop intelligent, high-performance unmanned aerial payload systems.

Skills & Tools

Programming languages: C++, C, Python, JavaScript, MATLAB, Assembly Language

Tools: TensorFlow, PyTorch, Scikit-learn, Deep Learning, Reinforcement Learning, CNNs, Transfer Learning, PID Optimization, Signal Processing, Transformers, Computer Vision, Vision Transformers (ViT), YOLOv8, Segment Anything (SAM), ONNX, TensorRT, RAG Pipelines, n8n

PROJECTS

Design Bio-feedback Integrated Wearable for Hand Tremor Reduction Using Reinforcement Learning (FYP)

- Designed a wearable device integrating gyroscopic stabilization and PID motor control to detect and counteract hand tremors in real-time. Implemented reinforcement learning for personalized tremor suppression, providing non-invasive and affordable solutions.

Explainable Skin Lesion Classification using Hybrid Learning Classifier Systems (In Progress)

- Developed and implemented a dermatological classification pipeline using the ExSTraCS learning classifier system and multi-modal polynomial feature interactions, achieving 72.4% balanced accuracy on external validation while demonstrating domain generalization.

Sign Language Recognition Using Image Processing and Machine Learning

- Developed a real-time Sign Language Recognition system using image processing and deep learning (CNN/ResNet50) to accurately classify hand gestures from live camera input.

Computer Vision Engineering Portfolio

- Designed and deployed robust algorithms for optic disc segmentation, feature-based camera localization, and optical flow motion tracking using both classical and deep learning methodologies.

CERTIFICATES

- IBM Data Science Professional Certificate
- Machine Learning Specialization, Stanford University